ITA0608 - Machine Learning
List of Lab Exercises

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim

To implement and demonstrate the **FIND-S algorithm** for finding the most specific hypothesis from a given set of training data.

Algorithm

1. Read the training dataset.
2. Initialize the hypothesis with the first positive example.
3. Compare the hypothesis with each positive training example.
4. If an attribute differs, replace it with `?`.
5. Ignore all negative examples.
6. Display the final most specific hypothesis.

Program

```
def find_s(dataset):

    """

    FIND-S Algorithm

    dataset = list of rows

    Last column = Target class

    Positive class = Yes / Positive

    """

    hypothesis = None

    for row in dataset:

        attributes = row[:-1]

        target = row[-1]

        if target in ["Yes", "Positive"]:

            if hypothesis is None:

                hypothesis = attributes.copy()
```

```
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = "?"
    return hypothesis
dataset1 = [
["Sunny","Warm","Normal","Strong","Warm","Same","Yes"],
["Sunny","Warm","High","Strong","Warm","Same","Yes"],
["Rainy","Cold","High","Strong","Warm","Change","No"],
["Sunny","Warm","High","Strong","Cool","Same","Yes"],
["Rainy","Warm","Normal","Weak","Warm","Same","No"],
["Sunny","Warm","Normal","Weak","Warm","Same","Yes"]
]
print("========== DATASET 1 : PLAY TENNIS ==========")
print("Final Hypothesis:", find_s(dataset1))
dataset2 = [
["High","Good","Permanent","Yes","Young","Yes"],
["High","Good","Permanent","No","Middle","Yes"],
["Low","Poor","Temporary","No","Young","No"],
["Medium","Good","Permanent","Yes","Middle","Yes"],
["High","Average","Temporary","Yes","Old","No"],
["High","Good","Permanent","Yes","Middle","Yes"]
]
print("\n========== DATASET 2 : LOAN APPROVAL ==========")
print("Final Hypothesis:", find_s(dataset2))
dataset3 = [
```

```
["High","Good","Yes","Good","High","Yes"],

["High","Excellent","Yes","Good","High","Yes"],

["Medium","Average","No","Average","Medium","No"],

["High","Good","Yes","Excellent","High","Yes"],

["Low","Poor","No","Average","Low","No"],

["High","Good","Yes","Good","Medium","Yes"]

]

print("\n========== DATASET 3 : STUDENT PLACEMENT ==========")

print("Final Hypothesis:", find_s(dataset3))

dataset4 = [

["Yes","Yes","Yes","Yes","Yes","Positive"],

["Yes","Yes","No","Yes","Yes","Positive"],

["No","Yes","Yes","No","No","Negative"],

["Yes","Yes","Yes","No","Yes","Positive"],

["No","No","Yes","Yes","No","Negative"],

["Yes","Yes","No","No","Yes","Positive"]

]

print("\n========== DATASET 4 : DISEASE DIAGNOSIS ==========")

print("Final Hypothesis:", find_s(dataset4))

dataset5 = [

["Yes","Yes","Yes","No","Yes","Yes"],

["Yes","Yes","Yes","Yes","Yes","Yes"],

["No","No","No","No","No","No"],

["Yes","No","Yes","No","Yes","Yes"],

["No","Yes","No","Yes","No","No"],

["Yes","Yes","Yes","No","No","Yes"]
```

```
]

print("\n========== DATASET 5 : EMAIL SPAM DETECTION ==========")

print("Final Hypothesis:", find_s(dataset5))
```

Output

```
["Yes","Yes","Yes","No","No","Yes"]
]
print("\n========== DATASET 5 : EMAIL SPAM DETECTION ==========")
print("Final Hypothesis:", find_s(dataset5))
```

```
========== DATASET 1 : PLAY TENNIS ==========
Final Hypothesis: ['Sunny', 'Warm', '?', '?', '?', 'Same']

========== DATASET 2 : LOAN APPROVAL ==========
Final Hypothesis: ['?', 'Good', 'Permanent', '?', '?']

========== DATASET 3 : STUDENT PLACEMENT ==========
Final Hypothesis: ['High', '?', 'Yes', '?', '?']

========== DATASET 4 : DISEASE DIAGNOSIS ==========
Final Hypothesis: ['Yes', 'Yes', '?', '?', 'Yes']

========== DATASET 5 : EMAIL SPAM DETECTION ==========
Final Hypothesis: ['Yes', '?', 'Yes', '?', '?']
```

Result

The FIND-S algorithm was successfully implemented. The algorithm found the most specific hypothesis that is consistent with all the positive training examples.

Final Hypothesis:['Sunny', 'Warm', '?', 'Strong', '?', '?']